

ATIVIDADE 11: SUB-REDES — PARTE 6

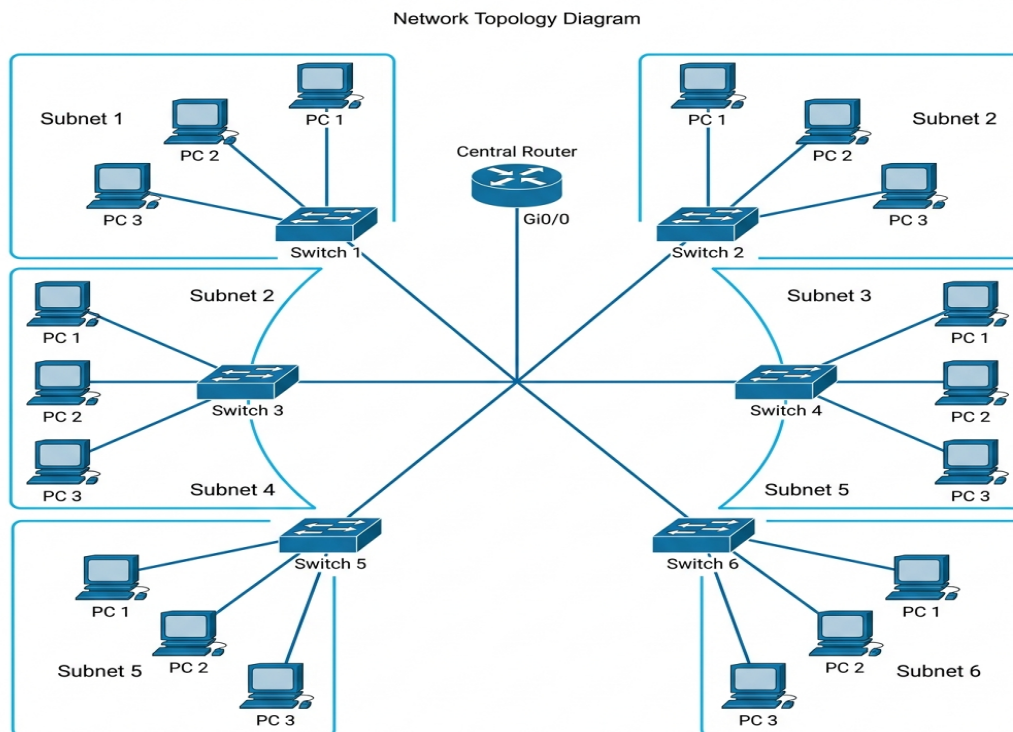
Empresa com 6 setores, 14 máquinas por setor. Rede Base Adotada: 192.168.10.0/24.

1. Memória de Cálculo

Para 14 hosts, calculamos $2^h - 2 \geq 14 \rightarrow h=4$ (14 hosts válidos). A máscara de rede passará a /28, com incrementos de bloco de 16 em 16. Provê 16 sub-redes.

Sub-rede	Endereço	Primeiro Válido	Último Válido	Broadcast
Setor 1	192.168.10.0/28	192.168.10.1	192.168.10.14	192.168.10.15
Setor 2	192.168.10.16/28	192.168.10.17	192.168.10.30	192.168.10.31
Setor 3	192.168.10.32/28	192.168.10.33	192.168.10.46	192.168.10.47
Setor 4	192.168.10.48/28	192.168.10.49	192.168.10.62	192.168.10.63
Setor 5	192.168.10.64/28	192.168.10.65	192.168.10.78	192.168.10.79
Setor 6	192.168.10.80/28	192.168.10.81	192.168.10.94	192.168.10.95

2. Topologia no Packet Tracer



3. Configurações de IP dos Hosts (1 por sub-rede)

Host Setor 1

IPv4 Address: 192.168.10.2
Subnet Mask: 255.255.255.240
Default Gateway: 192.168.10.1

Host Setor 2

IPv4 Address: 192.168.10.18
Subnet Mask: 255.255.255.240
Default Gateway: 192.168.10.17

Host Setor 3

IPv4 Address: 192.168.10.34
Subnet Mask: 255.255.255.240
Default Gateway: 192.168.10.33

Host Setor 4

IPv4 Address: 192.168.10.50
Subnet Mask: 255.255.255.240
Default Gateway: 192.168.10.49

Host Setor 5

IPv4 Address: 192.168.10.66
Subnet Mask: 255.255.255.240
Default Gateway: 192.168.10.65

Host Setor 6

IPv4 Address: 192.168.10.82
Subnet Mask: 255.255.255.240
Default Gateway: 192.168.10.81

4. Testes de Conectividade (Ping L3)

```
C:\>
C:\> ping 192.168.10.18
Pinging 192.168.10.18 with 32 bytes of data:
Reply from 192.168.10.18: bytes=32 time<1ms TTL=128
Reply from 192.168.10.18: bytes=32 time<1ms TTL=128
Reply from 192.168.10.18: bytes=32 time<1ms TTL=128
Reply from 192.168.10.18: bytes=32 time<1ms TTL=128

C:\> ping 192.168.10.34
Pinging 192.168.10.34 with 32 bytes of data:
Reply from 192.168.10.34: bytes=32 time<1ms TTL=128
Reply from 192.168.10.34: bytes=32 time<1ms TTL=128
Reply from 192.168.10.34: bytes=32 time<1ms TTL=128
Reply from 192.168.10.34: bytes=32 time<1ms TTL=128

C:\> ping 192.168.10.82
Pinging 192.168.10.82 with 32 bytes of data:
Reply from 192.168.10.82: bytes=32 time<1ms TTL=128
Reply from 192.168.10.82: bytes=32 time<1ms TTL=128
Reply from 192.168.10.82: bytes=32 time<1ms TTL=128
Reply from 192.168.10.82: bytes=32 time<1ms TTL=128
```